The vertical distance between the origin and the intersection point will be the Y coordinate of P. It will be denoted as P(y). The Y coordinate specifies how far the point is to the top or bottom from the origin along the Y axis. The intersection point of the horizontal and vertical lines is the coordinate of the point and is denoted as P(x,y). The X coordinate is positive, if measured from the right of the origin and is negative, if measured from the left of the origin. The Y coordinate is positive, if measured above the origin and is negative, if measured below the origin, refer to Figure 6.56.

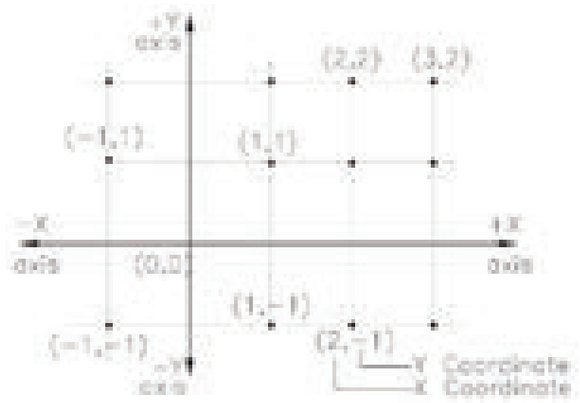


Fig. 6.56 Cartesian coordinate system

In AutoCAD, the default origin is located at the lower left corner of the drawing area.

AutoCAD uses the following coordinate systems to locate a point in the XY plane.

1. Absolute coordinates
2. Relative coordinates
3. Polar coordinates
4. Direct distance entry

If you are specifying a point by entering its location at the Command prompt then you need to use any one of the coordinate systems.

Absolute Coordinate System

In the Absolute Coordinate System, points are located with respect to the origin (0,0). For example, a point with X = 4 and Y = 3 is measured 4 units horizontally (distance along the X axis) and 3 units vertically (distance along the Y axis) from the origin, as shown in Figure 6.57. In AutoCAD, the absolute coordinates are specified at the Command prompt by entering X and Y coordinates, separated by a comma.

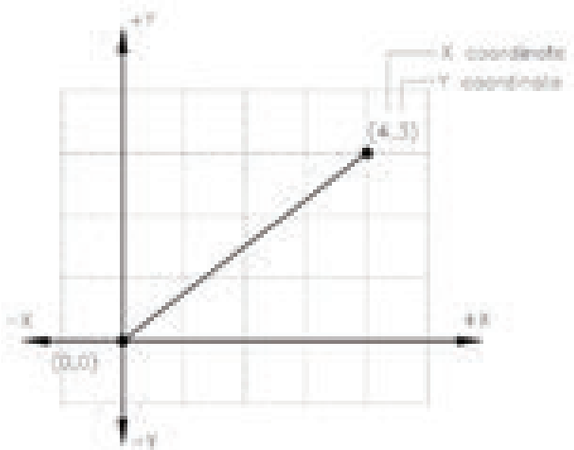


Fig. 6.57. Absolute Coordinate System

Draw a line on the points (50,50) and (150,50) using Absolute Co-ordinate System.

Command: LINE ↵

Specify first point: 50,50 ↵

Specify next point or [Undo]: 150,50 ↵

Specify next point or [Undo]: ↵ (Press Enter Key.)

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